Bitcoin Wallet Risk Analyzer

Project Status Report

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May 27, 2026

1. Is the development complete? Yes – the full system is working end to end. The offline side covers everything from preparing the data and building the wallet graphs, through training and calibrating the model, to producing an evaluation report and an importance score for each input feature. The user-facing side is a React web application backed by a FastAPI server that lets a visitor paste a Bitcoin address and receive a verdict, an explanation, and supporting transaction information.

2. What is missing? The main gaps are polish rather than core functionality: there is no automated test suite yet; the website is a single page and would benefit from a history view and a side-by-side comparison view; and the configuration that allows the production website to talk to the production API is still applied directly on the server rather than committed to the repository.

3. Is the system deployed? Yes. The website is hosted on the BGU CS department servers and is reachable at `https://cryptotrace.cs.bgu.ac.il/`. In addition to the web interface, we ship a Python (Jupyter / Google Colab) notebook that loads the trained model and walks through the same analysis on any address the reader supplies, so the model can be demonstrated end-to-end without needing access to our server.

4. Have you conducted experiments? Yes. We compared three models head to head, all judged on the same wallets and the same labels: (i) our main graph neural network, which reads both the wallet's own behaviour and the pattern of its transactions with its neighbours; (ii) an XGBoost classifier that sees only the wallet's own twelve summary statistics, with no neighbourhood information; and (iii) a simpler graph baseline that sees the neighbourhood but ignores the details of each transaction. Comparing the three lets us separate the contribution of each ingredient.

5. How were the experiments conducted? We combined two public datasets of labelled Bitcoin wallets, REAL-CATS (criminal wallets) and Elliptic++ (a mix of criminal and benign), removed overlap between them, and balanced the result so that exactly half of the wallets are criminal. This produced 86,870 wallets for training and 21,718 held out for testing. For every wallet we asked the public `mempool.space` service for its recent transactions and built a small graph around it: the wallet sits in the middle, the parties it transacted with sit around it, and each link records how much was sent, in which direction, and when. The main model was trained over multiple epochs with the usual safeguards (a validation split, early stopping, and a calibration step so the predicted probabilities can be read as honest confidence values). All three models were then evaluated on the same held-out wallets so the comparison is fair, and we produced the standard battery of diagnostic plots: confusion matrix, ROC curve, reliability diagram, and a per-feature importance score.

6. Important results. On the held-out test set:

Model	F_1	Accuracy	Recall	ROC-AUC
Our graph model	**0.856**	**0.913**	**0.897**	**0.961**
XGBoost (wallet stats only)	0.835	0.836	0.827	0.920
Simple graph baseline	0.633	0.761	0.674	0.809

Two gaps tell the story. The jump from the simple graph baseline to our model shows that the details of each transaction (amount, direction, timing) matter a great deal. The jump from XGBoost to our model shows that looking at the wallet's neighbourhood adds information that the wallet's own statistics do not capture. The calibration check came back essentially flat, so the confidence scores the website displays can be trusted at face value.

7. What is still lacking, and the schedule going forward. **Now – mid-June:** add a held-out adversarial test set of recent wallets in neither REAL-CATS nor Elliptic++ to stress-test generalisation. **Mid-June – end of June:** write the final report, polish the live demo, and build the presentation deck. **Early July:** project defence and final submission.